SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer **ALL** questions in this section. Marks are indicated at the end of each question. Together they are worth 80% of the total marks for this examination.

Question 1 (20 marks – approximately 36 minutes)

Kimberlite Company Limited ("Kimberlite") is a company listed on the Hong Kong Stock Exchange. The information on the capital structure of Kimberlite is as follows:

Common stock: Total number of shares outstanding is 10 million with the total book value of HK$100 million. The current share price is HK$13 per share. The beta coefficient is 0.85.

Preferred stock: Total number of shares outstanding is 1 million with par value of HK$100 per share. The dividend rate is 5% per annum. The current selling price is HK$96 per share.

Bonds: Total number of units outstanding is 120,000 with an annual coupon rate of 6% and face value of HK$1,000 per unit. The current selling price is HK$1,000 per unit.

Other relevant information:
Expected return on the market is 12%.

The rate of three-month Hong Kong exchange fund bills is 3%.

Tax rate is 16.5%.

Required:

(a) **Calculate the cost of common stock, cost of preferred stock and pre-tax cost of debt for Kimberlite.**

(5 marks)

(b) **Calculate the weighted average cost of capital ("WACC") for Kimberlite.**

(4 marks)

(c) **Explain the "Efficient Market Hypothesis", list its THREE forms of market efficiency, and describe each of them respectively.**

(7 marks)

(d) **The management of Kimberlite is working on a restructuring plan which will have a major impact on the profitability of the company. Assume a semi-strong efficient capital market, analyse the change in share price of Kimberlite under the following circumstances:**

(i) **The restructuring plan has just been approved by the board of directors of Kimberlite but it has not been announced to the public yet.**

(2 marks)

(ii) **The restructuring plan has just been announced to the public.**

(2 marks)

Question 2 **(20 marks – approximately 36 minutes)**

Rocky is a qualified accountant and lives in a village house in the New Territories of Hong Kong. Recently, the power supply company launched a solar energy promotional scheme ("SEPS") with the aim of encouraging the development of renewable solar energy by allowing residential customers to sell the electricity generated from the solar energy systems ("SES") back to the power supply company at favourable tariff rates. Under the SEPS, the power supply company will enter into a five-year contract with the residential customers who installed the SES and purchase the electricity at a rate of HK$4 per unit while the current tariff rates that it charges the residential customers are less than HK$2 per unit. Hematite Consultancy Limited ("Hematite") specialises in consultancy services relating to the construction of the SES for the SEPS. Hematite approached Rocky and suggested two proposals for him to consider:

Proposal A:
An initial investment of HK$100,000 is required for the construction of the SES. The SES will have a life span of five-years with no residual value. It is estimated that there would be an on-going annual maintenance cost of HK$2,000 payable to Hematite.

Proposal B:
Rocky does not need to pay for the initial investment and annual maintenance cost. However, Rocky needs to pay Hematite a fee of HK$2.2 for each unit of electricity generated during the five-year contract period with the power supply company.

Other relevant information:
The required rate of return for Hematite's capital is 12%.

The forecasted electricity generated by the SES is 12,000 units per annum for the next five years.

The rate of the three-month Hong Kong exchange fund bills is 3%.

Required:

(a) Considering the risk level of this project, Rocky estimated the required rate of return for this project should be 2% over the risk-free rate. From Rocky's perspective, calculate the net present values for the TWO proposals suggested by Hematite respectively and determine which proposal should be accepted.

(7 marks)

(b) From Hematite's perspective, the cost of constructing the SES is HK$70,000 and the annual maintenance cost is HK1,500. Hematite will adopt the straight-line depreciation method for the SES which will have no residual value after the end of the 5-year contract period. For Proposal B, calculate the accounting rate of return ("ARR") from Hematite's perspective and determine whether it is viable for Hematite if Rocky chooses this proposal.

(7 marks)

(c) From the power supply company's perspective, explain THREE non-financial considerations why the power supply company would like to launch this SEPS which would have a negative impact to its profitability.

(6 marks)

<u>Question 3</u> **(20 marks – approximately 36 minutes)**

Andesite Company Limited ("Andesite") is a privately owned company selling fast-moving consumer goods on its own online trading platform. The revenues of Andesite have grown rapidly in the past two years due to the pandemic which changed the shopping behaviours of citizens. Andesite currently has now come to the stage of requires additional capital for expansion.

The management of Andesite is considering purchasing automatic stock picking equipment and delivery trucks which can improve the operational efficiency and accuracy of delivery services. The estimated cost of this equipment is HK$20 million. In a board meeting of Andesite, the Chief Financial Officer ("CFO") suggested that Andesite should approach Vivianite for the required capital of HK$20 million. Vivianite is a venture capitalist which provides finance to small businesses with high growth potential. Alternatively, the Chief Executive Officer ("CEO") suggested Andesite should contact the bank which is providing the existing loan for an increase of the loan amount from HK$60 million to HK$80 million.

Other relevant information:
The net income from the latest Statement of Profit or Loss for the year ended 31 December 2021 is HK$15 million.

Dividend payout ratios for the last three years are 35% and the management of Andesite has no intention to change this payout ratio. Andesite has just paid the dividend for the year ended 31 December 2021.

Estimated growth rate in earnings is 10% per annum in perpetuity.

Total number of ordinary shares outstanding is 50 million.

Existing bank loan outstanding is HK$60 million.

The cost of equity for Andesite is 13%.

Tax rate is 16.5%.

Average P/E ratio for listed companies similar to Andesite is 14. A discount of 20% should be applied to estimate the P/E ratio for Andesite as a private company’s shares are not publicly available for purchase and sales.

Required:

(a) **The CFO understands that Vivianite will normally require to purchase the shares of Andesite in order to provide financing. Calculate the numbers of new shares that Andesite will have to issue and the percentage of ownership acquired by Vivianite in order to obtain the HK$20 million from Vivianite using the following valuation methods:**

(i) **Dividend valuation model; and**

(4 marks)

(ii) **Price/ earnings ratio method.**

(4 marks)

(b) **Explain the operation of a "venture capitalist" for providing funding and identify the probable tactics employed by it to mitigate risk.**

(6 marks)

(c) **For Andesite's circumstances, discuss TWO advantages and TWO disadvantages of obtaining the additional capital from a venture capitalist as compared to obtaining a long-term loan from a bank.**

(6 marks)

Question 4 (20 marks – approximately 36 minutes)

Zoisite Limited ("Zoisite") is a trading company selling paper packaging products for the food and beverage industry. Zoisite has just released the financial results for the year ended 31 December 2021.

Extracts from the Statement of Profit or Loss for the year ended 31 December 2021

	HK$'000
Net sales	8,050
Cost of goods sold	5,400
Gross profit	2,650
Selling and general administrative expenses	1,687
Profit before interest and taxes	963
Interest expenses	245
Profit before tax	718
Tax expenses	118
Net profit	600

Extracts from the Statement of Financial Position as at 31 December

	2021	2020
	HK$'000	HK$'000
Non-current assets	4,405	4,150
Current assets		
Inventories	1,600	1,230
Accounts receivable	1,580	1,280
Cash	2,235	2,520
	5,415	5,030
Total assets	9,820	9,180
Non-current liabilities	3,000	3,000
Current liabilities		
Accounts payable	1,290	1,250
	1,290	1,250
Share capital	1,200	1,200
Retained earnings	4,330	3,730
Total equity	5,530	4,930
Total liabilities and equity	9,820	9,180

David is the newly appointed CFO of Zoisite. He just obtained some industry average statistics on working capital management of Zoisite's industry. The industry averages were as follows:

Working capital management ratios	**Industry averages**
Days' Inventory	65
Days' Receivables	55
Days' Payables	60

Required:

(a) (i) **For the year ended 31 December 2021, based on the available financial information of Zoisitie, calculate the THREE working capital management ratios (in days) with industry averages obtained by the CFO for Zoisite. For your calculations, the assumption of a 365-day year and average balances instead of the ending balances of the items from the statement of financial position should be used. Your answers should be rounded to the nearest whole number.**

(6 marks)

(ii) **For the year ended 31 December 2021, calculate the cash conversion cycles ("CCC") for industry average and Zoisite respectively. Your answers should be rounded to the nearest whole number.**

(2 marks)

(iii) **Based on the results in Question 4(a)(i) and 4(a)(ii) above, analyse Zoisite's working capital management efficiency as compared to the industry averages, and recommend how to improve if necessary.**

(6 marks)

(b) **Define the concept of net working capital. Explain the importance and objectives of working capital management of a company.**

(6 marks)

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